

Octopus Deploy

The Power of GitOps and the Rise of ArgoCD



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Welcome agenda

1. Introduction to GitOps
2. Scope and lack of standards
3. Promotion to different envs
4. Observability & Traceability
5. Mixing CI and CD
6. GitOps at scale
7. The future of GitOps
8. The Rise of ArgoCD
9. Argo CD main features
10. ArgoCD Demo

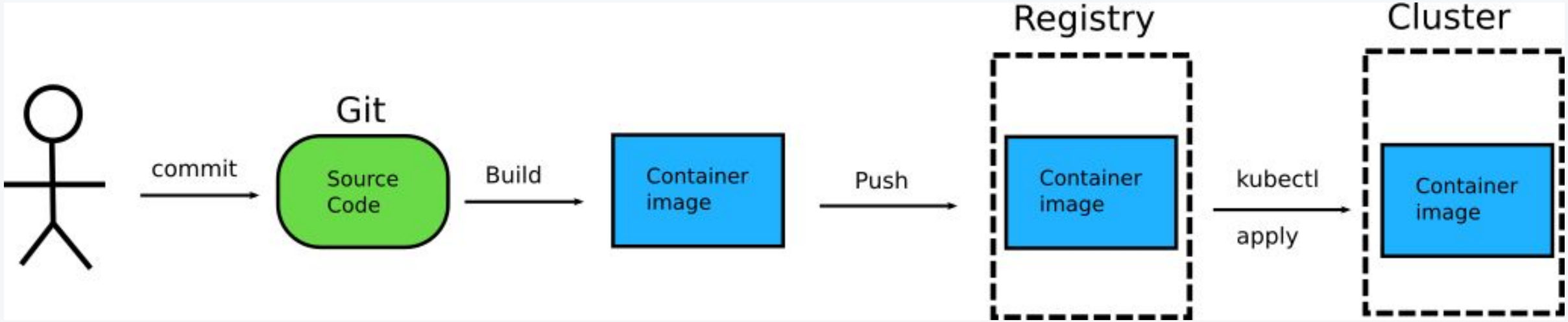


What is GitOps

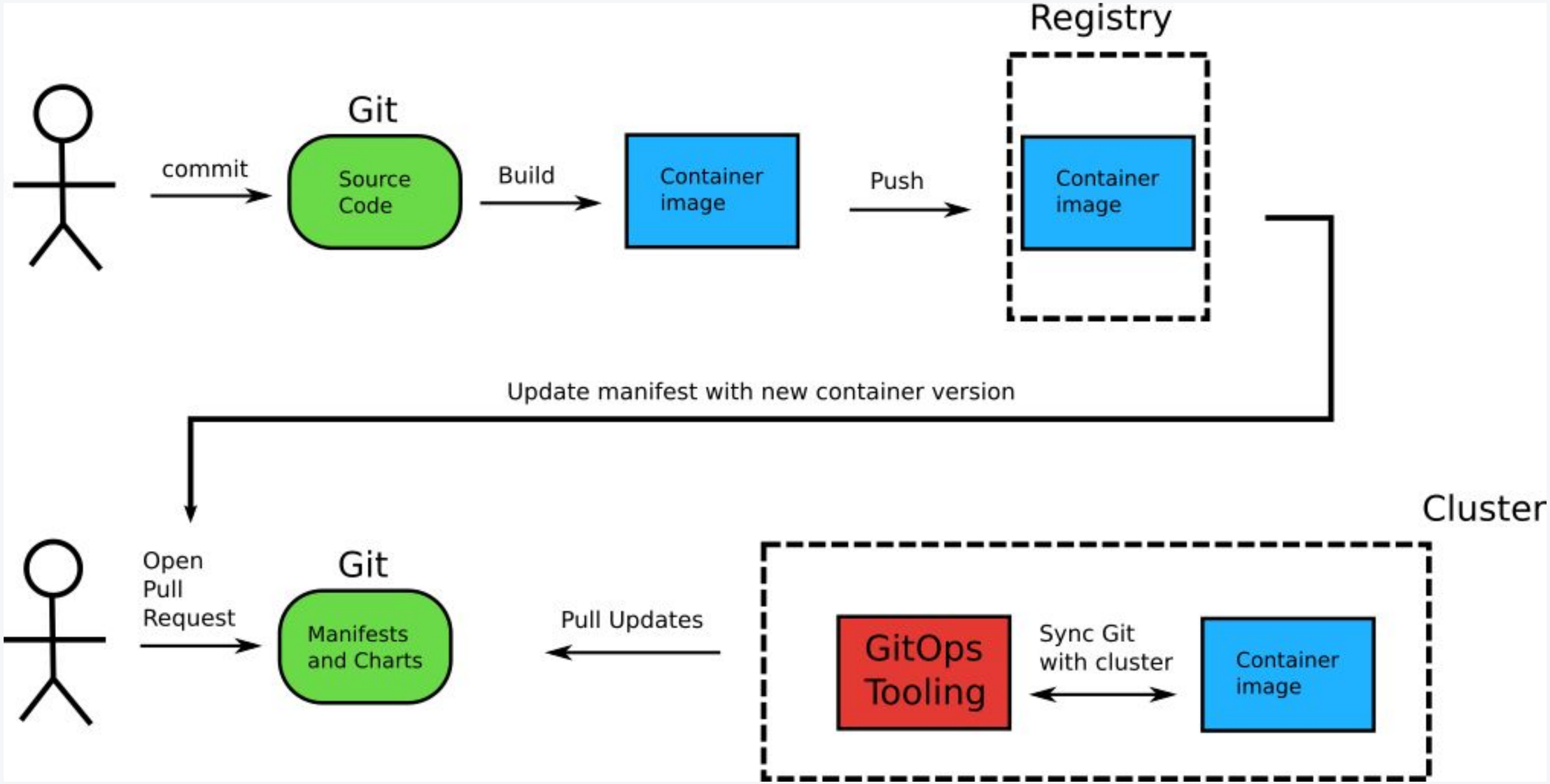
- Everything is Git
- Declarative Deployments
- No Ad-hoc operations



Traditional Deployment



GitOps Deployment



GitOps Advantages

- Git state = cluster state
- Git History = Deployment History
- Deploy / Rollback with Git tools

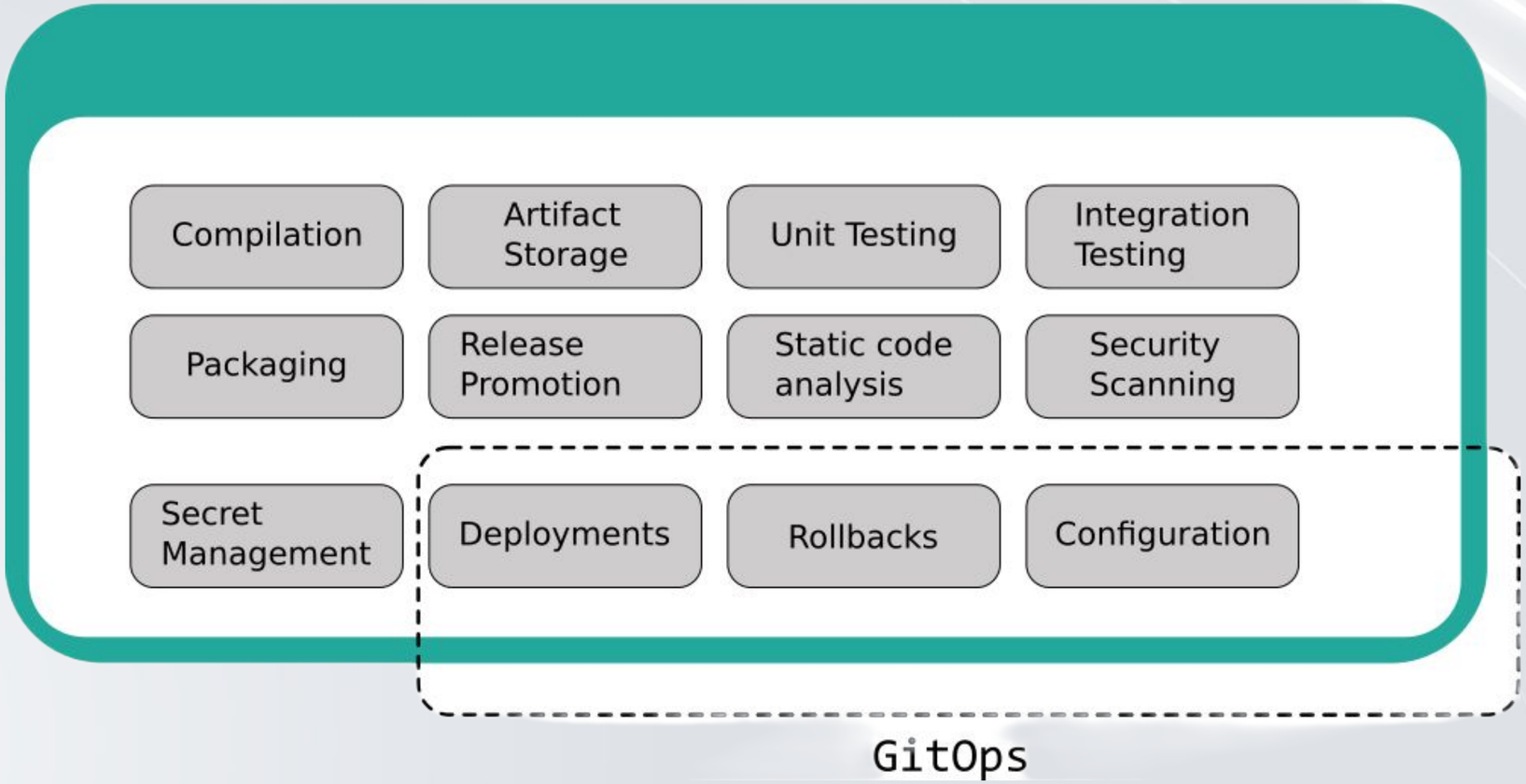


GitOps Challenges

- Scope
- How to properly do rollbacks?
- Secrets management



GitOps only covers deployments





Environment promotions

Tacticus 11 points · 1 year ago

How do you handle promotion through different environments or rollback in the event of a bad service deploy?

Give Award Share Report Save

coderanger 3 points · 1 year ago

Branches, different envs are configured to watch different branches of the gitops repos. Promotion is opening a PR to merge from one to the next. Rollback is undoing that merge.

Give Award Share Report Save

Tacticus 1 point · 1 year ago

So you don't have any automated progression through environments. manual branch merges are it?

Give Award Share Report Save

GassiestFunInTheWest 1 point · 1 year ago

Do you run multiple environments? How do you orchestrate releases through and across environments?

Give Award Share Report Save

Tacticus 7 points · 1 year ago

Gitops tools really need to find a better way of doing multi environment promotion.

Give Award Share Report Save

GassiestFunInTheWest 3 points · 1 year ago

Agreed. Without this, GitOps is strictly worse than CD pipelines. From what I can see though, it's not something that the GitOps movement is trying to address.

As such, I'm going to stick with push-based CD pipelines.

Give Award Share Report Save



First Environment

Commit to App repo

Create image

Push to registry

Open / Merge PR to manifest repo

Sync Cluster



Second environment?



Observability
Typical Cluster questions

- Is the application healthy?
- How many replicas are running?
- Is traffic routing correct?
- When was the last deployment?
- Is Git state same as cluster state?



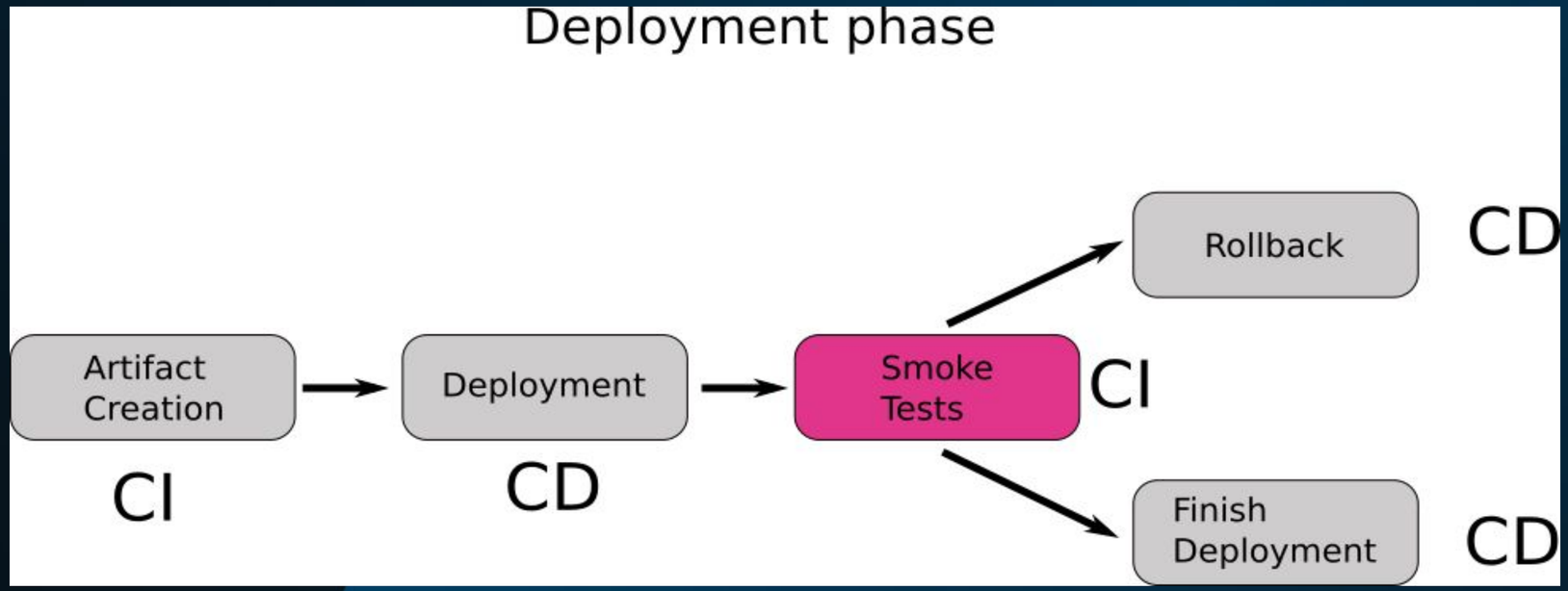
Observability
Typical application questions

- Is feature X shipped to prod?
- Is bug Y present on staging?
- What features did we deploy this week?
- How long did we let staging bake till we deployed to prod?



Mixing CI and CD

- How to do smoke tests?
- How to do parallel deployments?
- Pre and post deployment steps





GitOps at Scale

- How do we handle 10 environments?
  - 10 git repos?
  - Multiple parent-child repos?
  - 1 git repo with 10 folders?
  - 1 git repo with 10 branches?
- Other problems
  - How to find common settings across all environments?
  - How to promote releases from one env to another?
  - How to coordinate deployments?
  - How to handle auditing?



The Rise of ArgoCD



Get More Done with Kubernetes

Open source tools for Kubernetes to run workflows, manage clusters, and do GitOps right.

View on GitHub



Trusted by

Adobe NVIDIA TESLA Google Red Hat WordPress ticketmaster

https://argoproj.github.io/

Argo CD

16997

Declarative continuous delivery with a fully-loaded UI.

Learn More





Argo Workflows

14685

Kubernetes-native workflow engine supporting DAG and step-based workflows.

Learn More



Argo Rollouts

2619

Advanced Kubernetes deployment strategies such as Canary and Blue-Green made easy.

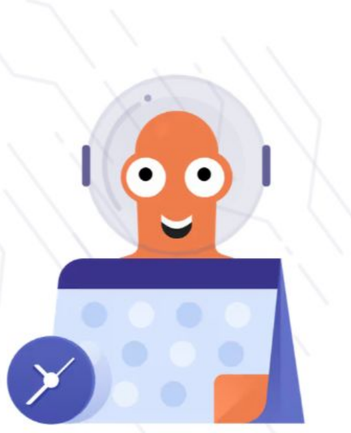
Learn More

Argo Events

2299

Event based dependency management for Kubernetes.

Learn More



CNCF End User Survey Finds Argo CD as Majority Adopted GitOps Solution for Kubernetes

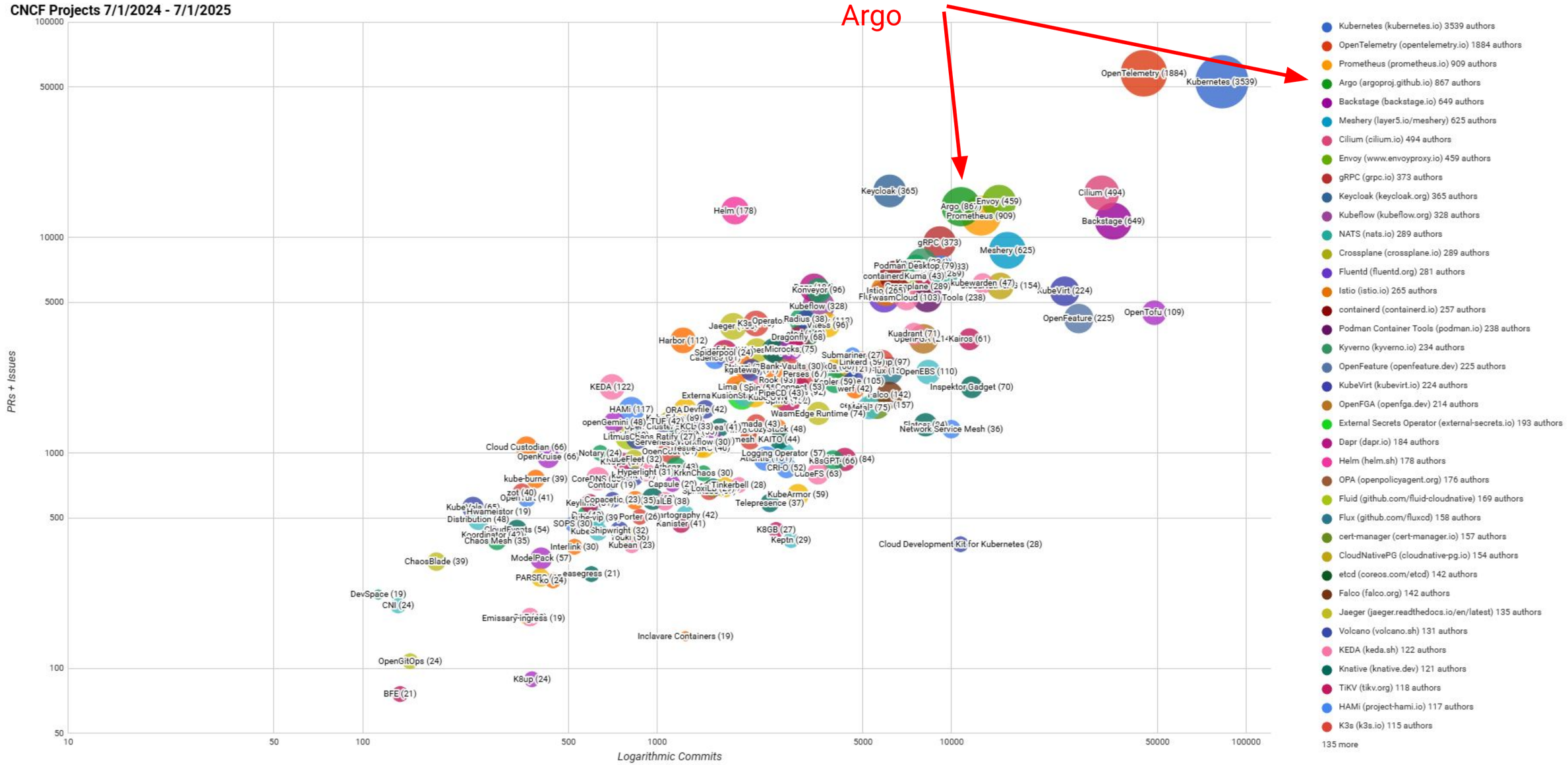


Nearly 60% of Kubernetes clusters managed by survey respondents now rely on Argo CD, with strong satisfaction fueled by 3.0 performance and security updates

https://www.cncf.io/announcements/2025/07/24/cncf-end-user-survey-finds-argo-cd-as-majority-adopted-gitops-solution-for-kubernetes/



Popular/Active CNCF projects

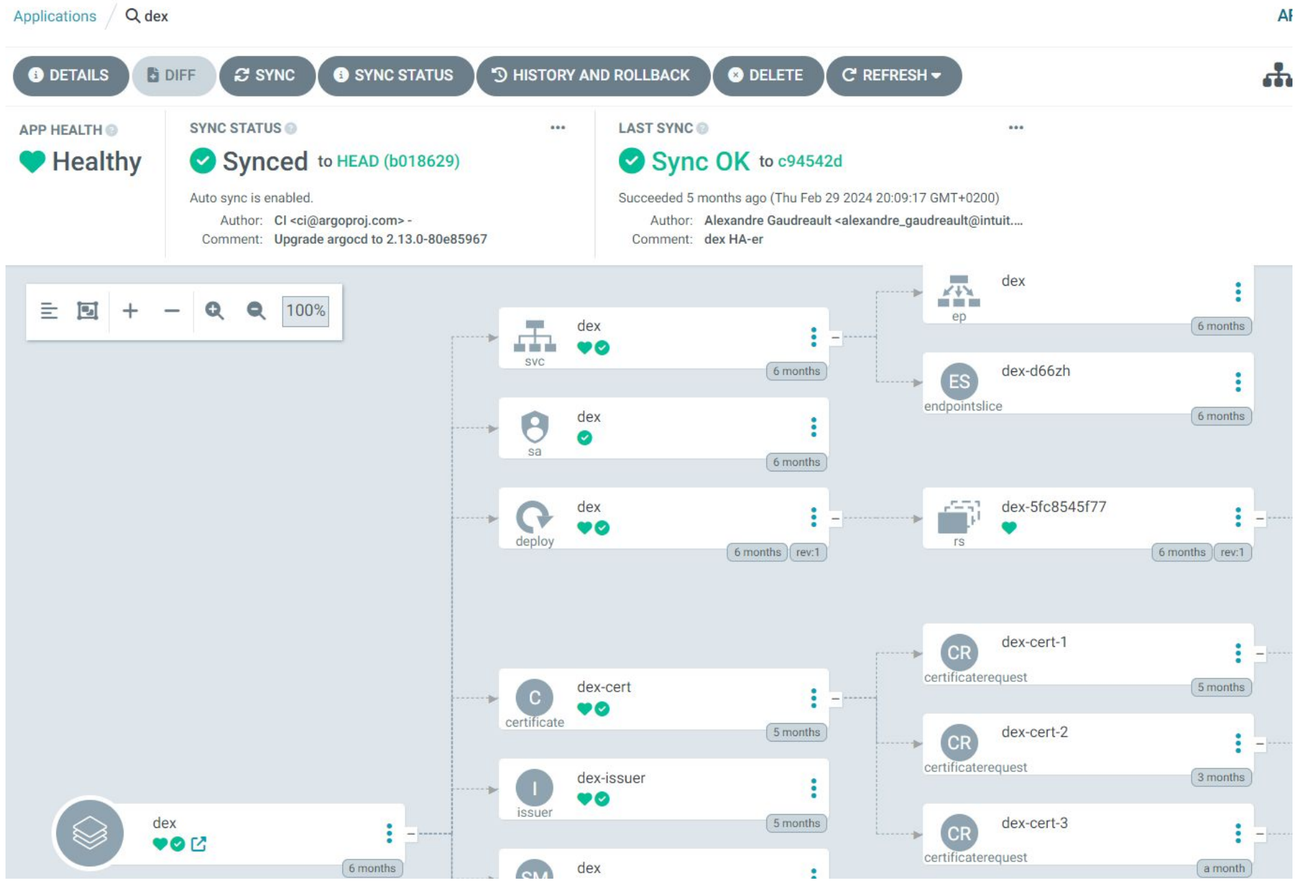


Argo CD

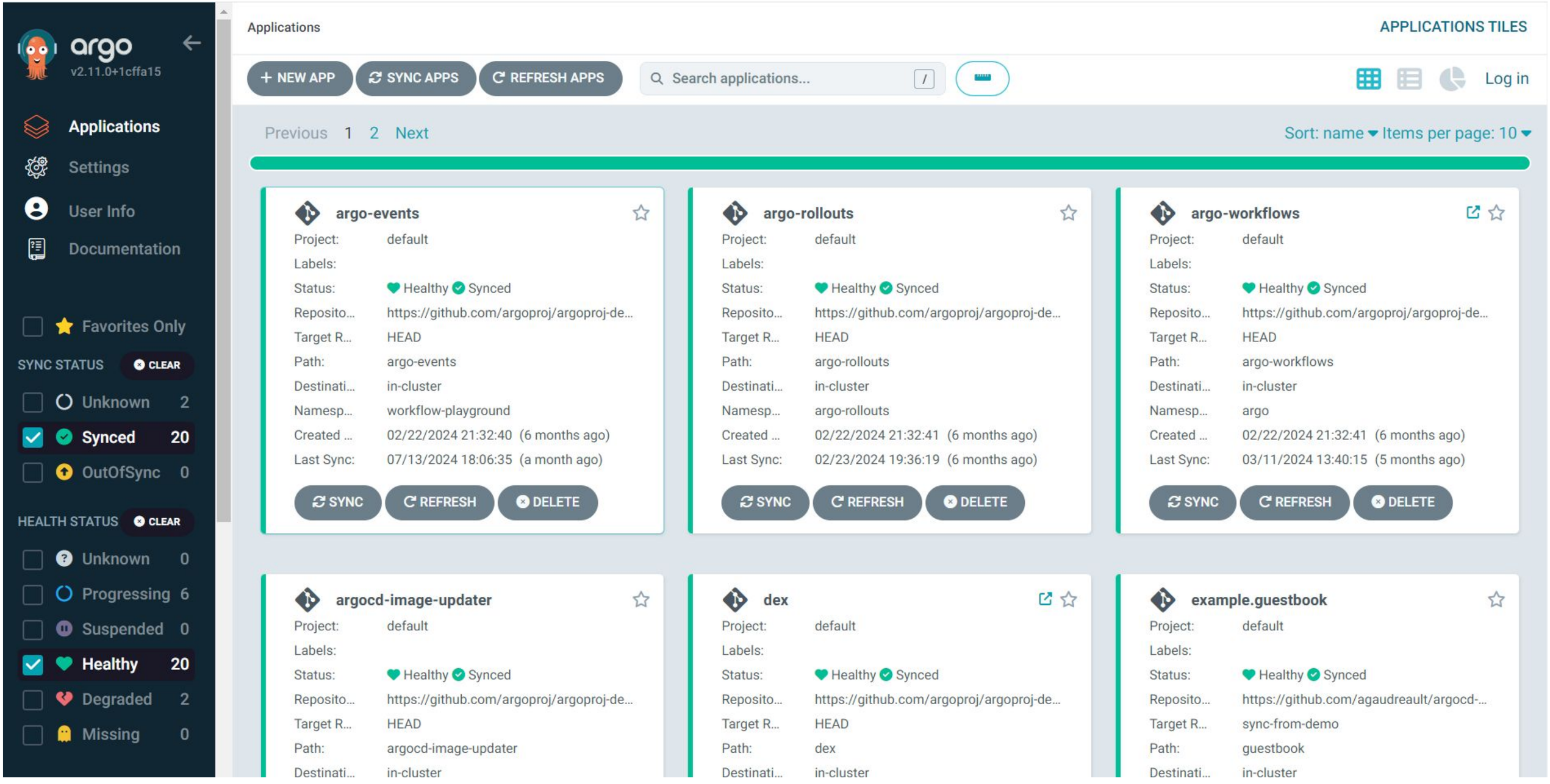


Argo CD

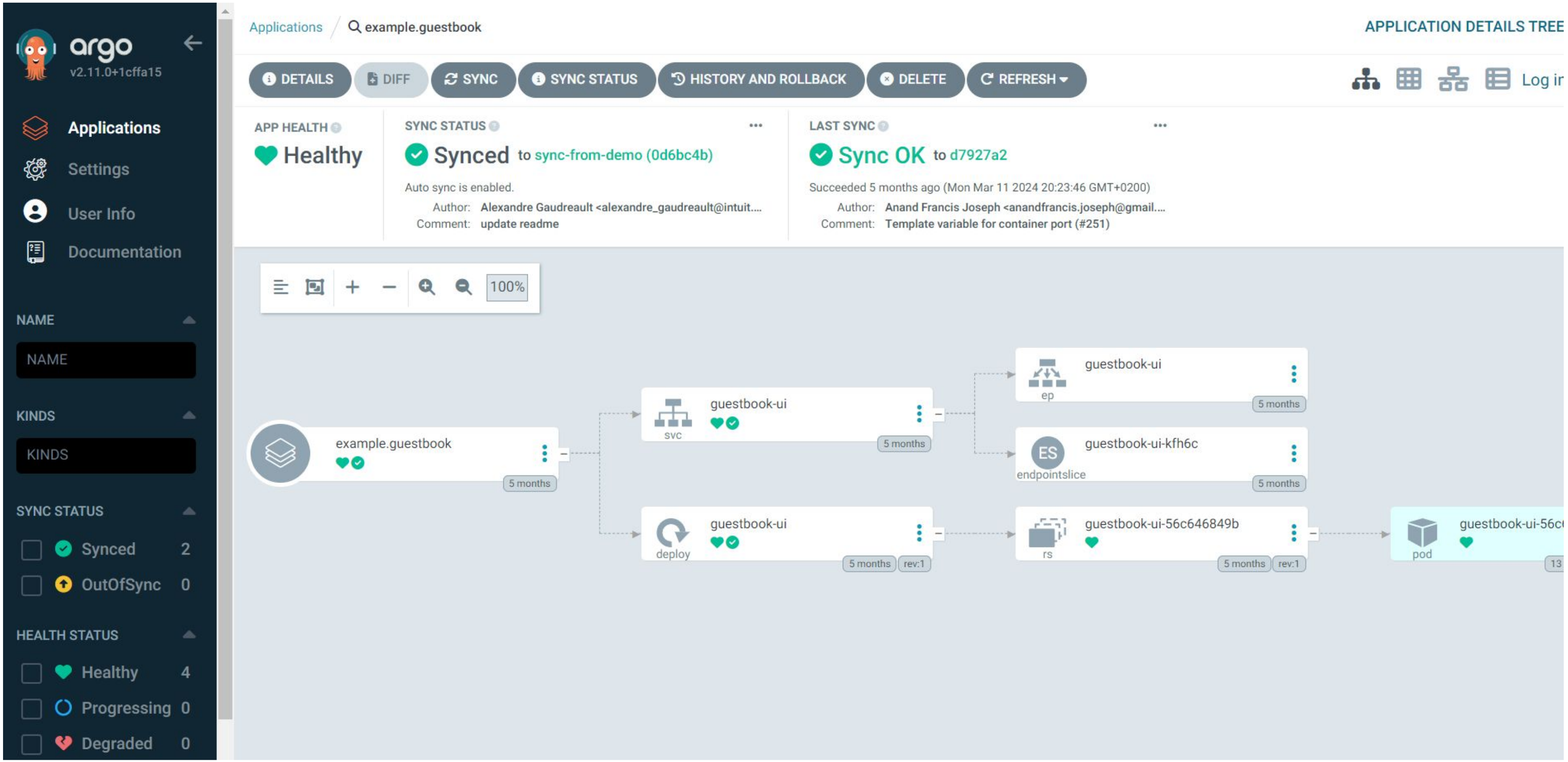
- Deploys applications
- Kubernetes native
- Supports Helm/Kustomize
- Health status analysis
- Multi-tenant/RBAC

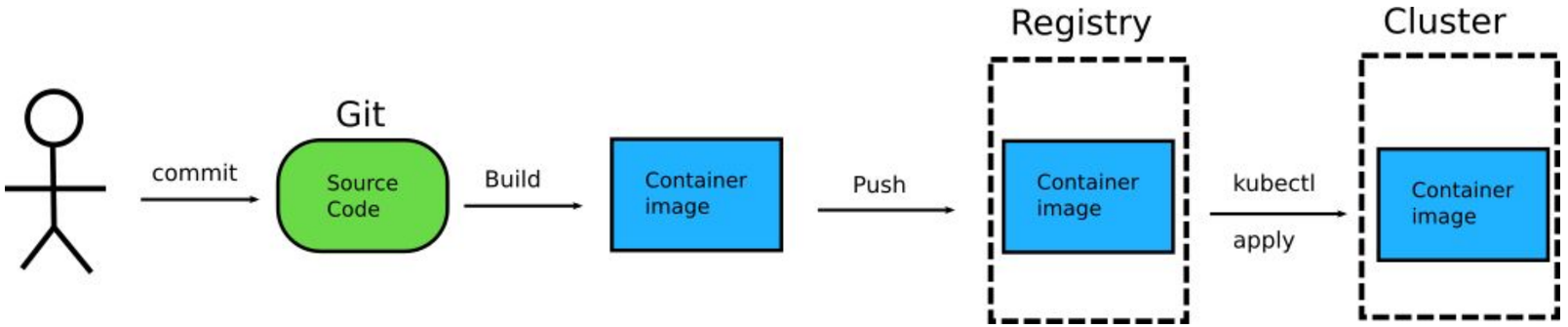


Argo CD UI

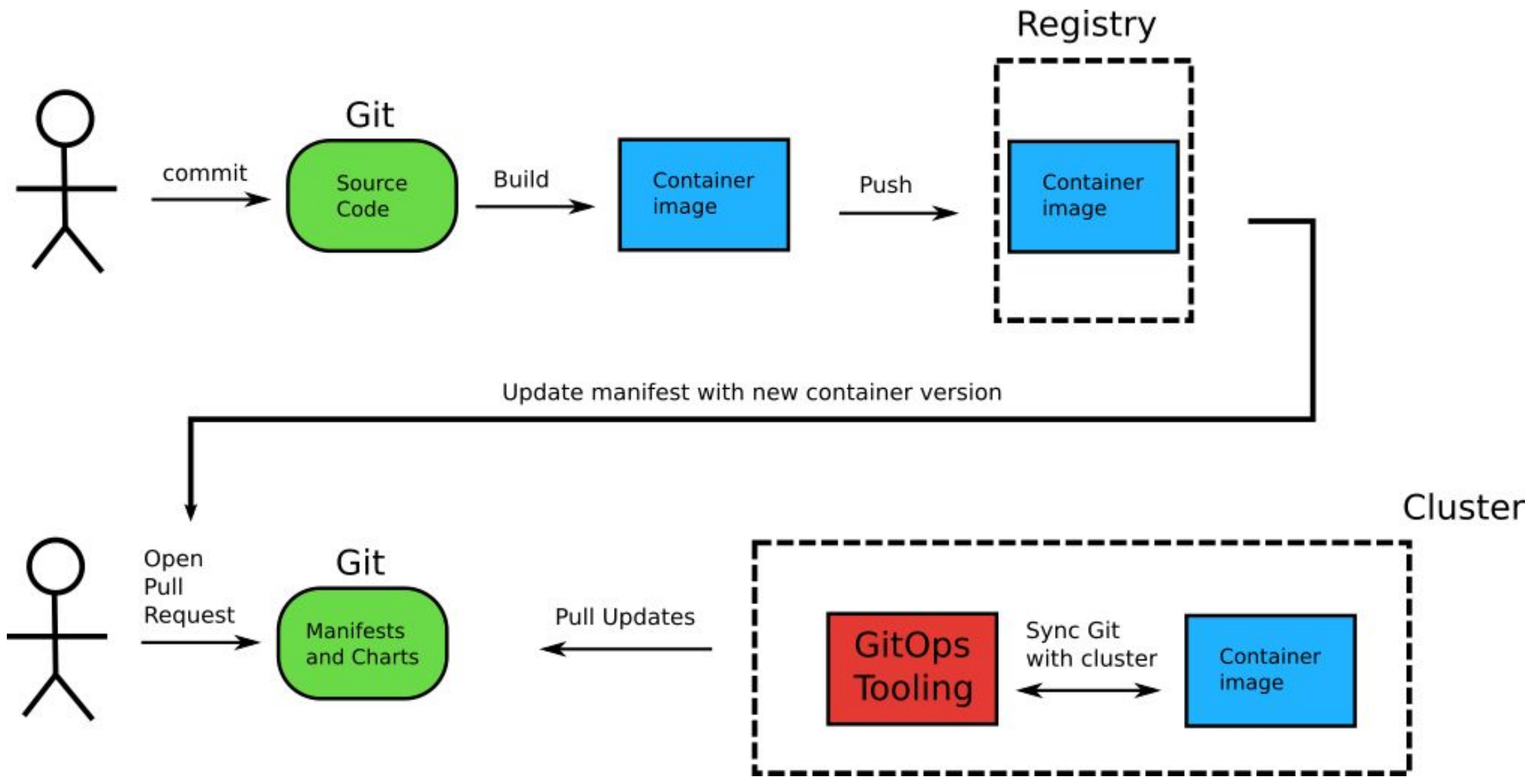

argo
v2.11.0+1cffa15
Applications
Settings
User Info
Documentation
Favorites Only
SYNC STATUS
CLEAR
Unknown 2
Synced 20
OutOfSync 0
HEALTH STATUS
CLEAR
Unknown 0
Progressing 6
Suspended 0
Healthy 20
Degraded 2
Missing 0
Applications
APPLICATIONS TILES
+ NEW APP
SYNC APPS
REFRESH APPS
Search applications...
/
Log in
Previous 1 2 Next
Sort: name
Items per page: 10
argo-events
Project: default
Labels:
Status: Healthy Synced
Reposito... https://github.com/argoproj/argoproj-de...
Target R... HEAD
Path: argo-events
Destinati... in-cluster
Namesp... workflow-playground
Created ... 02/22/2024 21:32:40 (6 months ago)
Last Sync: 07/13/2024 18:06:35 (a month ago)
SYNC
REFRESH
DELETE
argo-rollouts
Project: default
Labels:
Status: Healthy Synced
Reposito... https://github.com/argoproj/argoproj-de...
Target R... HEAD
Path: argo-rollouts
Destinati... in-cluster
Namesp... argo-rollouts
Created ... 02/22/2024 21:32:41 (6 months ago)
Last Sync: 02/23/2024 19:36:19 (6 months ago)
SYNC
REFRESH
DELETE
argo-workflows
Project: default
Labels:
Status: Healthy Synced
Reposito... https://github.com/argoproj/argoproj-de...
Target R... HEAD
Path: argo-workflows
Destinati... in-cluster
Namesp... argo
Created ... 02/22/2024 21:32:41 (6 months ago)
Last Sync: 03/11/2024 13:40:15 (5 months ago)
SYNC
REFRESH
DELETE
argocd-image-updater
Project: default
Labels:
Status: Healthy Synced
Reposito... https://github.com/argoproj/argoproj-de...
Target R... HEAD
Path: argocd-image-updater
dex
Project: default
Labels:
Status: Healthy Synced
Reposito... https://github.com/argoproj/argoproj-de...
Target R... HEAD
Path: dex
example.guestbook
Project: default
Labels:
Status: Healthy Synced
Reposito... https://github.com/agaudreault/argocd-...
Target R... sync-from-demo
Path: guestbook

Argo CD UI


argo
v2.11.0+1cffa15
Applications
Settings
User Info
Documentation
NAME
NAME
KINDS
KINDS
SYNC STATUS
Synced 2
OutOfSync 0
HEALTH STATUS
Healthy 4
Progressing 0
Degraded 0
Applications
example.guestbook
APPLICATION DETAILS TREE
DETAILS
DIFF
SYNC
SYNC STATUS
HISTORY AND ROLLBACK
DELETE
REFRESH
Log in
APP HEALTH
Healthy
SYNC STATUS
Synced to sync-from-demo (0d6bc4b)
Auto sync is enabled.
Author: Alexandre Gaudreault <alexandre_gaudreault@intuit....
Comment: update readme
LAST SYNC
Sync OK to d7927a2
Succeeded 5 months ago (Mon Mar 11 2024 20:23:46 GMT+0200)
Author: Anand Francis Joseph <anandfrancis.joseph@gmail....
Comment: Template variable for container port (#251)
100%
example.guestbook
5 months
guestbook-ui
svc
5 months
guestbook-ui
ep
5 months
guestbook-ui-kfh6c
ES
endpointslice
5 months
guestbook-ui
deploy
5 months
rev:1
guestbook-ui-56c646849b
rs
5 months
rev:1
pod



Abusing CI as CD



With Argo CD



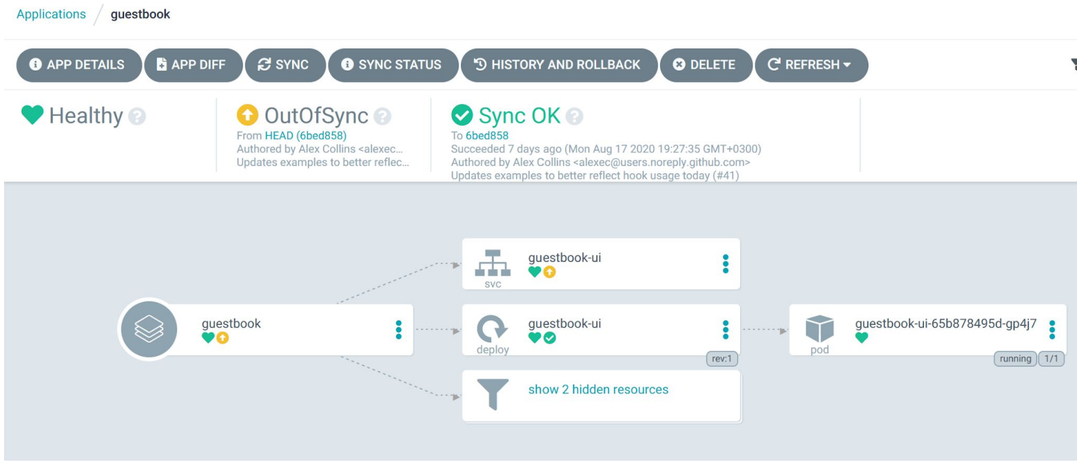
Avoid Configuration Drift

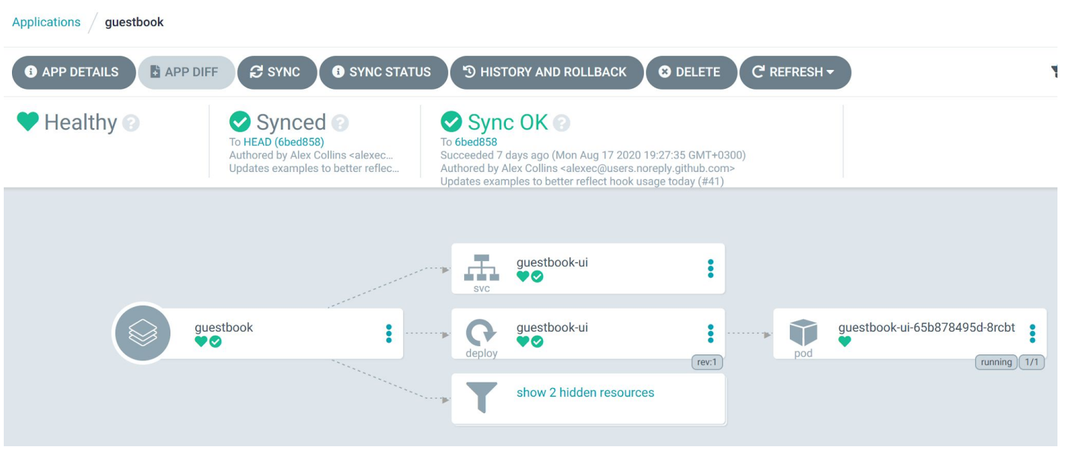
SUMMARY | PARAMETERS | MANIFEST | DIFF | EVENTS

Compact diff | Inline Diff

/Service/default/guestbook-ui

```
 1  apiVersion: v1                                  1  apiVersion: v1
 2  kind: Service                                   2  kind: Service
 3  metadata:                                       3  metadata:
 4    labels:                                       4    labels:
 5      app.kubernetes.io/instance: guestbook       5      app.kubernetes.io/instance: guestbook
 6    name: guestbook-ui                            6    name: guestbook-ui
 7  spec:                                           7  spec:
 8    ports:                                        8    ports:
                                                    9      - port: 80
                                                   10        targetPort: 80
 9      - port: 8080                               11      - port: 8080
10        targetPort: 80                           12        targetPort: 80
11    selector:                                    13    selector:
12      app: guestbook-ui                          14      app: guestbook-ui
```

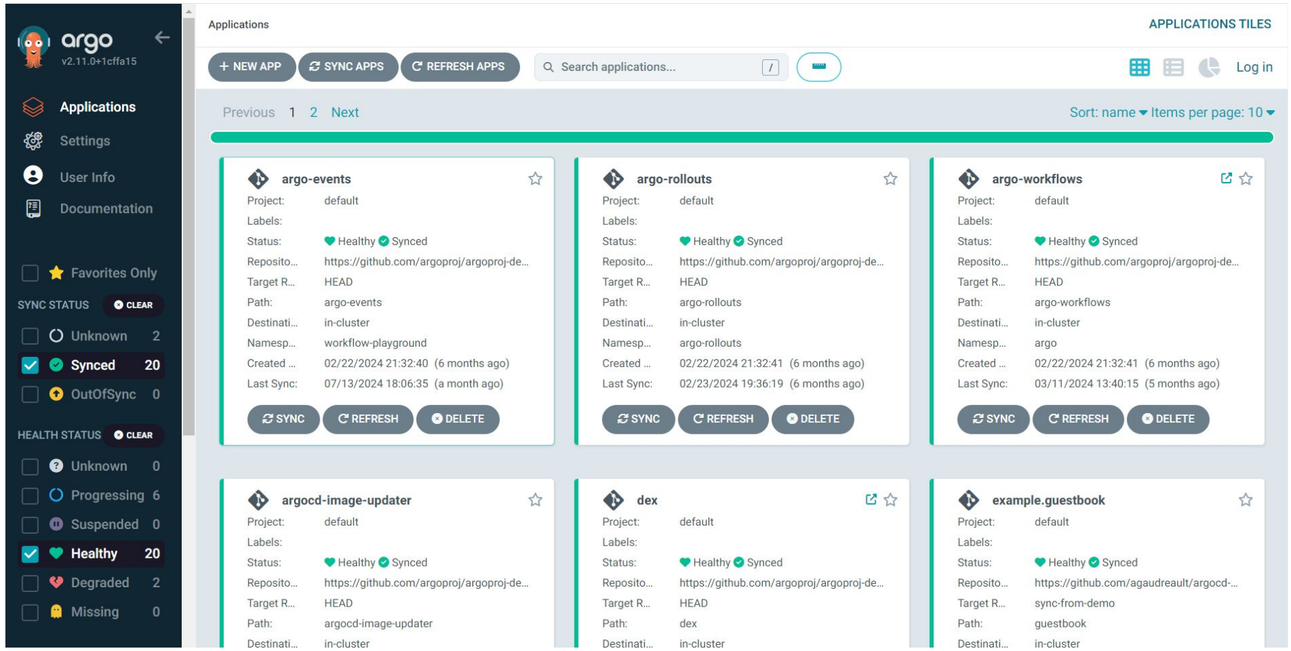






Argo CD entities

- **Application**- Link between a cluster and Git repo
- **Project** - RBAC for Applications
- **ApplicationSet**- Generator/grouping for applications





Sync manifests to Cluster

```yaml
apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: guestbook
  namespace: argocd
spec:
  project: default
  source:
    repoURL: https://github.com/argoproj/argocd-example-apps.git
    targetRevision: HEAD
    path: guestbook
  destination:
    server: https://kubernetes.default.svc
    namespace: guestbook
```

Name of application

Where to read the Kubernetes manifest

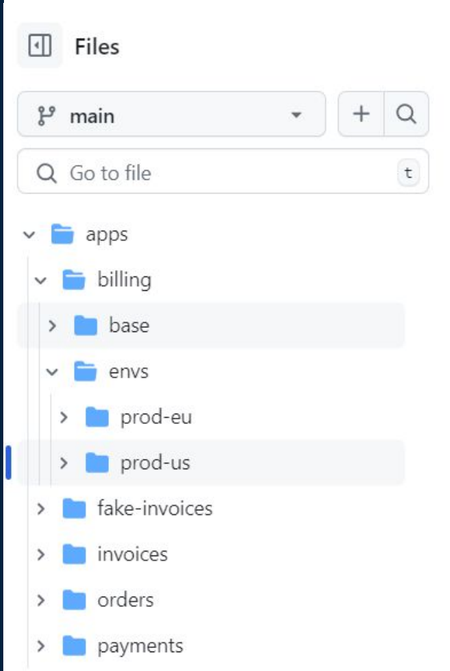
Which cluster to deploy the application to

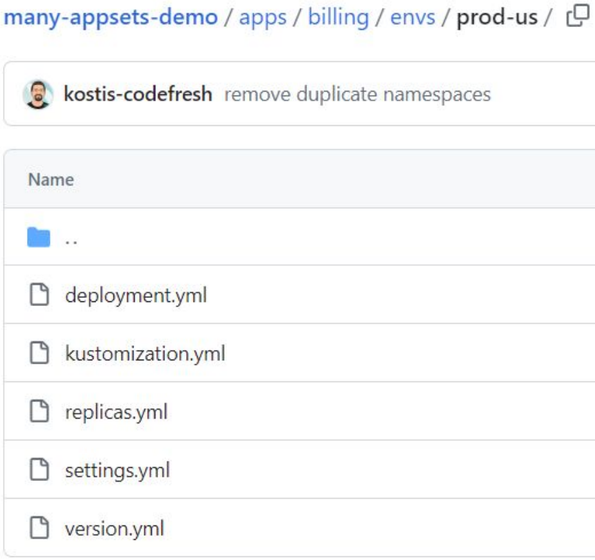


```yaml
apiVersion: argoproj.io/v1alpha1
kind: ApplicationSet
metadata:
  name: my-qa-appset
  namespace: argocd
spec:
  goTemplate: true
  goTemplateOptions: ["missingkey=error"]
  generators:
  - git:
    repoURL: https://github.com/kostis-codefresh/many-appsets-demo.git
    revision: HEAD
    directories:
    - path: apps/*/envs/qa
  template:
  metadata:
    name: '{{index .path.segments 1}}-{{index .path.segments 3}}'
  spec:
    # The project the application belongs to.
    project: default

    # Source of the application manifests
    source:
      repoURL: https://github.com/kostis-codefresh/many-appsets-demo.git
      targetRevision: HEAD
      path: '{{.path.path}}'

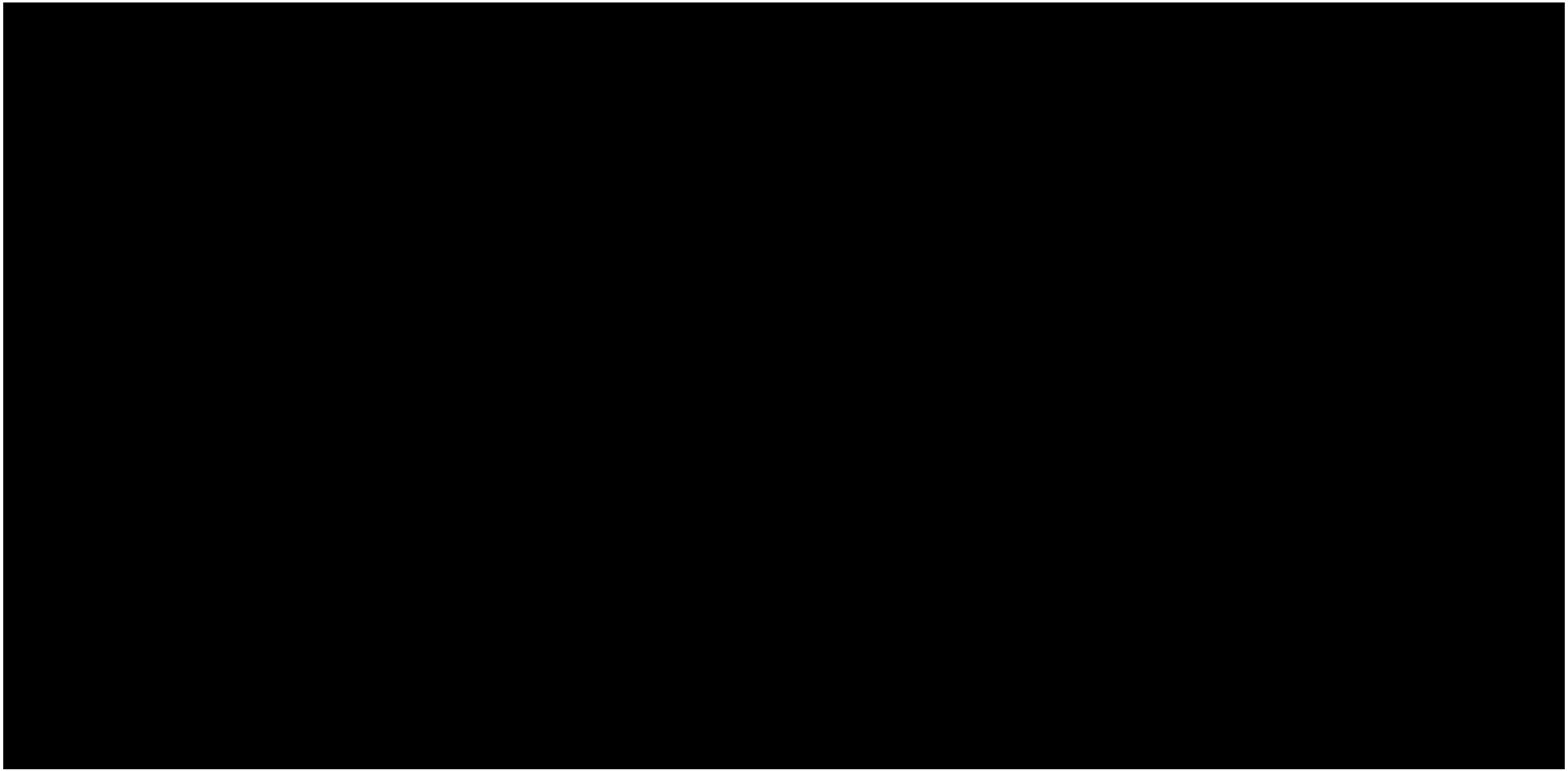
    # Destination cluster and namespace to deploy the application
    destination:
      server: https://kubernetes.default.svc
      namespace: '{{index .path.segments 1}}-{{index .path.segments 3}}'
```





Generate applications from Git folders

Cluster bootstrapping



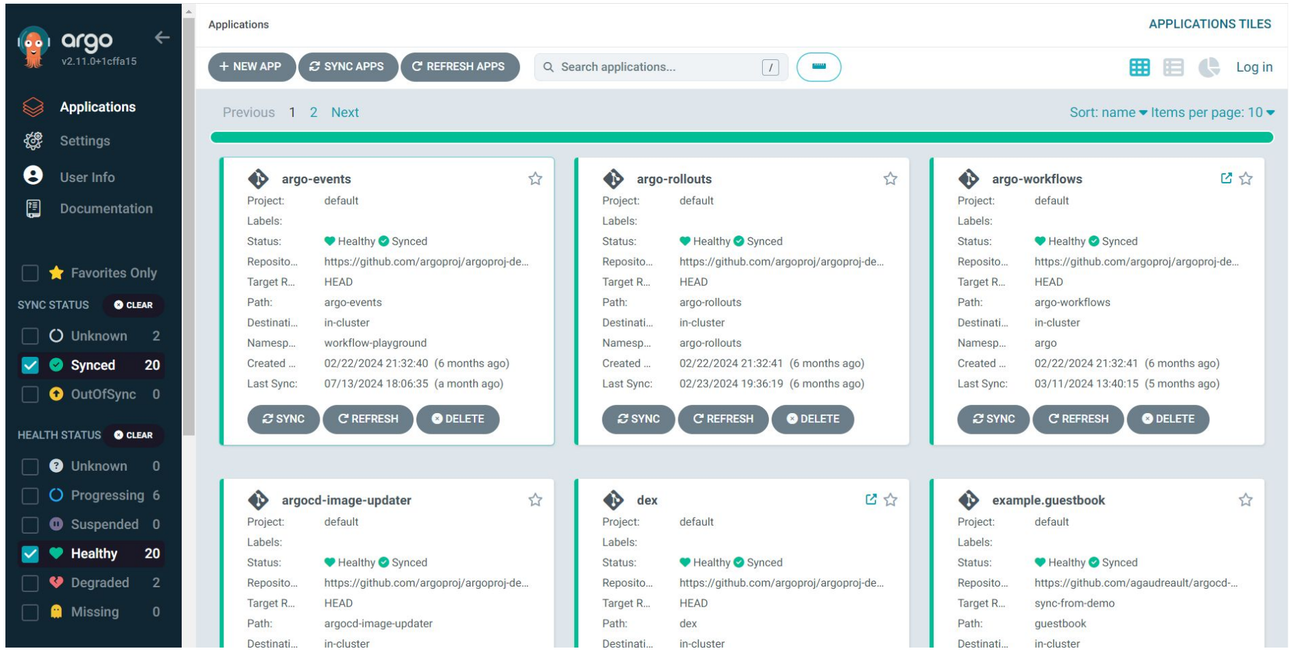
Argo CD topologies





Argo CD - other features

- Sync policies
- Sync waves/phases/windows
- Git webhooks
- CLI/API
- SSO/RBAC
- OCI Support
- Plugins
- Notifications





Thank you

Discussion / Questions

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